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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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Authors: Matthijs Smakman, Ana Tanevska

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Project Coordinator: Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

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Introduction

1.1 SymAware: High-level description of the project

SymAware addresses the fundamental need for a new conceptual framework for awareness in multi- agent systems (MASs) that is compatible with the internal models and specifications of robotic agents and that enables safe simultaneous operation of collaborating autonomous agents and hu- mans. The goal of SymAware is to provide a comprehensive framework for situational awareness to support sustainable autonomy via agents that actively perceive risks and collaborate with other robots and humans to improve their awareness and understanding while fulfilling complex and dynamically changing tasks. The SymAware framework will use compositional logic, symbolic computations, formal reasoning, and uncertainty quantification to characterise and support situational awareness of MAS in its various dimensions, sustaining awareness by learning in social contexts, quantifying risks based on limited knowledge, and formulating risk-aware negotiation of task distributions. These objectives will be achieved in SymAware through

1. logical characterization of awareness using symbolic methods,
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware address the “Awareness Inside” Challenge of the European Innovation Council (EIC) by extending and formalizing human-based models of situational awareness and by providing a novel conceptual situational awareness framework for MASs that encompasses logical characterisation and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal, risk and social dimensions. This will support transitioning to safe mixed operation of autonomous agents and humans.

Report on potential misuse of research

Report by dr. Matthijs Smakman, independent ethics advisor on ethical aspects and potential misuse of research of the SYMAWARE project

The SYMAWARE project aims to create a novel conceptual framework for situational awareness in multi-agent systems that is compatible with the internal models and specifications of robot agents and that enables the safe simultaneous operation of collaborating autonomous agents and humans. The project is mainly conceptual with a high level of technical complexity with limited involvement of human participants. The envisioned end result of the project will be a virtual environment wherein autonomous vehicles and drones can navigate.

To identify and mitigate any potential misuse of results and ethical issues the SYMAWARE project appointed an external ethical advisor dr. Matthijs Smakman, Associate Professor HU University of Applied Sciences Utrecht (author of this report).

All participating groups of the SYMAWARE project filled out a questionnaire designed to elicit potential misuse of research. In 2023 all six participating groups (MPI-SWS, TU/e, KTH, UU, NLR and SISW) had completed the questionnaire: None of the groups could envision that the materials, methods, technologies and knowledge involved or generated could harm humans, animals or the environment if they are modified or enhanced. Also, they indicated that the materials, methods, technologies and knowledge involved or generated could not realistically serve any other purposes than the intended application(s). Furthermore, no unethical potential misuse was indicated by the groups.

Next to the questionnaire, a workshop has been organized with all involved groups in June 2023 in Eindhoven, the Netherlands to discuss the potential misuse of research. The workshop resulted in no realistic potential misuse of materials, methods, technologies and knowledge related to the SYMAWARE project. The main relevant concerns related to the potential misuse of the results, such as the risks of misuse with multi-agent systems – if implemented by people who wish to harm, who might modify the risk calculation formula to select options with maximum risk, but this was considered a very unlikely scenario.

In conclusion, I see no major risks in the potential misuse of the materials, methods, technologies and knowledge related to the SYMAWARE project. To ensure that any new potential misuse of research is identified and mitigated during the project, the workshop and questionnaire should have a yearly follow-up.

Sincerely,

dr. Matthijs Smakman
HU University of Applied Sciences